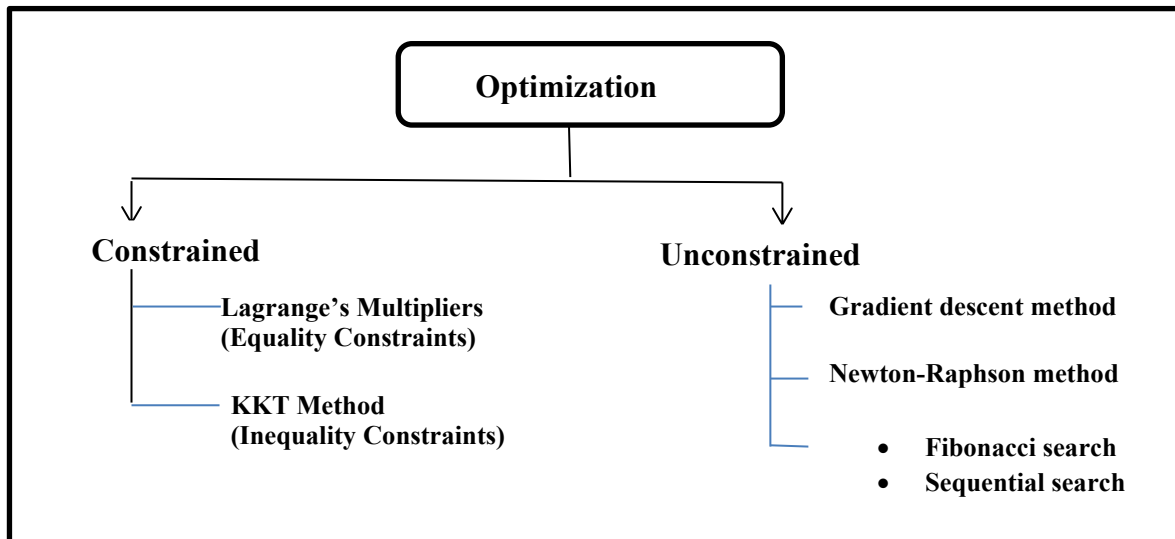




Unit 2: Continuous Optimisation-2



Constrained Optimization

Lagrange Multipliers:

Property: The gradient of a function is perpendicular to the contour lines, the tangents to the contour lines of f and g are parallel if and only if the gradients of f and g are parallel.

Thus we want points (x, y) where $g(x, y) = c$ and $\nabla f - \sum \lambda_i \nabla g_i = 0$ where λ is unrestricted sign.

Method of Lagrange multipliers (Equality constraints)

Given $u=f(x, y, z)$ subject to $g(x, y, z) = 0$

Step-1: Write the Auxiliary Equation $F(x, y, z) = f(x, y, z) - \lambda g(x, y, z)$

Step-2: Find F_x , F_y and F_z by differentiating F partially with respect to 'x', 'y' and 'z' respectively.

Step-3: Solve the equations $F_x = 0$, $F_y = 0$ and $F_z = 0$ together with $\phi(x, y, z) = 0$.

Step-4: The values of x, y and z so obtained will give the stationary value of $f(x, y, z)$.

Note: Multiple constraints problem:

$u=f(x, y, z)$ Subject to $\phi(x, y, z) = 0$, and $\psi(x, y, z) = 0$ then

Auxiliary equation $F(x, y, z) = f(x, y, z) - \lambda_1 \phi(x, y, z) - \lambda_2 \psi(x, y, z)$.

Follow the same steps as we defined above.

Problems:

- Find the maximum and minimum distances of the point $(3, 4, 12)$ from the sphere $x^2 + y^2 + z^2 = 1$.
- Find the maximum and minimum value of the function $f(x, y) = 5x - 3y$ subject to the constraint $x^2 + y^2 = 136$.



3. Find the maximum and minimum values of $f(x, y, z) = 3x^2 + y$ subject to the constraints $4x - 3y = 9$ and $x^2 + z^2 = 9$.
4. Find the maximum and minimum values of $f(x, y, z) = 4y - 2z$ subject to the constraints $2x - y - z = 2$ and $x^2 + y^2 = 1$.
5. Find the maximum and minimum values of $f(x, y, z) = 2x + 4y + z^2$ subject to the constraints $y^2 + z^2 = 1$ and $x^2 + z^2 = 1$.
6. Find the maximum and minimum values of $f(x, y, z) = x + y + z^2$ subject to the constraints $x + y + z = 1$ and $x^2 + z^2 = 1$.
7. Find the extreme values of the function $f(x, y, z) = 2x + 3y + z$ subject to the constraints $x^2 + y^2 = 5$ and $x + z = 1$.
8. An asteroid is entering the atmosphere of moon. The shape of the asteroid is described by the equation $4x^2 + y^2 + 4z^2 = 16$. The temperature on the surface of the asteroid after one month was observed to be represented by equation $8x^2 + 4yz - 16z + 600$. Is it possible to find the point on surface of asteroid with maximum temperature? If yes, find it?

Karush-Kuhn-Tucker (KKT) optimality conditions: (Inequality constraints)

For the problem: Min $f(x)$ subject to $g(x) \leq b$ (n variables, m constraints) The necessary conditions are: $\nabla f - \sum \lambda_i \nabla g_i = 0$ (optimality) $g_i(x) - b_i \leq 0$ for $i = 1, 2, \dots, m$ (feasibility) $\lambda_i(g_i(x) - b_i) = 0$ for $i = 1, 2, \dots, m$ (Complementary slackness condition) $\lambda_i \leq 0$ for $i = 1, 2, \dots, m$ (non-positivity)	For the problem: Max $f(x)$ subject to $g(x) \leq b$ (n variables, m constraints) The necessary conditions are: $\nabla f - \sum \lambda_i \nabla g_i = 0$ (optimality) $g_i(x) - b_i \leq 0$ for $i = 1, 2, \dots, m$ (feasibility) $\lambda_i(g_i(x) - b_i) = 0$ for $i = 1, 2, \dots, m$ (Complementary slackness condition) $\lambda_i \geq 0$ for $i = 1, 2, \dots, m$ (non-negativity)
---	---

Note: The non-positivity or non-negativity of λ_i depends on the way the optimality condition is written.

Maximize $f(x)$

s.t. $g_i(x) \leq b_i$

the **Necessary conditions**

$$\frac{\partial f}{\partial x} - \sum_i \lambda_i \frac{\partial g_i}{\partial x} = 0$$

$$\lambda_i(g_i(x) - b_i) = 0$$

$$g_i(x) \leq b_i$$

$$\lambda_i \geq 0$$

Minimize $f(x)$

s.t. $g_i(x) \leq b_i$

the **Necessary conditions**

$$\frac{\partial f}{\partial x} - \sum_i \lambda_i \frac{\partial g_i}{\partial x} = 0$$

$$\lambda_i(g_i(x) - b_i) = 0$$

$$g_i(x) \leq b_i$$

$$\lambda_i \leq 0$$



For NLPPs

Case - I Maximize $f(x)$ s.t. $g_i(x) \leq b_i$	Case - III Minimize $f(x)$ s.t. $g_i(x) \leq b_i$	Case I: f is CONCAVE and g is CONVEX Case II: f is CONCAVE and g is CONCAVE Case III: f is CONVEX and g is CONVEX Case IV: f is CONVEX and g is CONCAVE
Case - II Maximize $f(x)$ s.t. $g_i(x) \geq b_i$	Case - IV Minimize $f(x)$ s.t. $g_i(x) \geq b_i$	

Problems:

1. Maximize $Z = 12x_1 + 21x_2 + 2x_1x_2 - 2x_1^2 - 2x_2^2$ subject to the constraints $x_1 + x_2 \leq 10$; $x_2 \leq 8$ and $x_1, x_2 \geq 0$ by deriving the KKT conditions.
2. Minimize $f = x_1^2 + 2x_2^2 + 3x_3^2$ subject to $x_1 - x_2 - 2x_3 \leq 12$, $x_1 + 2x_2 - 3x_3 \leq 8$ by deriving the KKT conditions.
3. Minimize $f = x_1^2 + x_2^2 + 60x_1$ subject to $x_1 \geq 80$, $x_1 + x_2 \geq 120$ by deriving the KKT conditions.
4. Maximize $f(x, y) = xy$ subject to the constraints $x + y^2 \leq 2$; $x, y \geq 0$ by deriving the KKT conditions.
5. Maximize x subject to: $y \leq (1 - x)^3$ and $y \geq 0$ by deriving the KKT conditions.
6. Maximize $-(x - 2)^2 - 2(y - 1)^2$ subject to $x + 4y \leq 3$ and $x \geq y$ by deriving the KKT conditions.
7. Minimize $(x - 3)^2 + (y - 2)^2$ subject to: $x^2 + y^2 \leq 5$, $x + 2y \leq 4$ and $x, y \geq 0$ by deriving the KKT conditions.

Unconstrained Optimization**1. Newton's Method to optimize quadratic functions**

consider the quadratic approximation of the function $f(X)$ at $X = X_i$ using the Taylor's series expansion

$$f(X) = f(X_i) + [\nabla f_i](X - X_i) + \frac{1}{2}(X - X_i)^T [H_i](X - X_i) \dots (1),$$

where $[\nabla f_i]$ is $\left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \quad \frac{\partial f}{\partial x_3} \dots \right]$ and $[H_i]$ is the Hessian matrix of f evaluated at the point X_i .

By setting the partial derivatives to zero for the minimum of $f(X)$, we obtain

$$\frac{\partial f(X)}{\partial x_j} = 0, j = 1, 2 \dots n \dots (2)$$

From 1 and 2 give

$$[\nabla f]^T = [\nabla f_i]^T + [H_i](X - X_i) = 0 \dots (3)$$

If $[H_i]$ is non-singular, equations (3) can be solved to obtain an improved approximation ($X = X_{i+1}$) as

$$X = X_i - [H_i]^{-1} \cdot [\nabla f_i]^T \dots (4)$$

Since higher-order terms have been neglected in Eq. (1), Eq. (4) is to be used iteratively to find the optimum solution X^* .



Note: If $f(X)$ is a non-quadratic function, Newton's method may sometimes diverge, and it may converge to saddle points and relative maxima. This problem can be avoided by modifying Eq. (4) as

$$X_{i+1} = X_i + \lambda_i^* S_i = X_i - \lambda_i^* [H_i]^{-1} \cdot \nabla f_i^T \dots (5)$$

The modification indicated by Eq. (5) has a number of advantages.

- 1) it will find the minimum in lesser number of steps compared to the original method.
- 2) it finds the minimum point in all cases, whereas the original method may not converge in some cases.
- 3) it usually avoids convergence to a saddle point or a maximum.

With all these advantages, this method appears to be the most powerful minimization method.

Despite these advantages, the method is not very useful in practice, due to the following

- 1) It requires the storing of the $n \times n$ matrix $[H_i]$.
- 2) It becomes very difficult and sometimes impossible to compute the elements of the matrix $[H_i]$.
- 3) It requires the inversion of the matrix $[H_i]$ at each step.
- 4) It requires the evaluation of the quantity $[H_i]^{-1} \cdot \nabla f_i$ at each step.

Problems:

1. Minimize $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$ by taking the starting point as $X_1 = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$.
2. Minimize $f(x_1, x_2) = x_1^2 + 4x_2^2$ by taking the starting point as $X_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$.
3. Minimize $f(x) = x_1^2 + x_2^2 + x_1x_2 + 10(x_1 + x_2)$ by taking the starting point as $X_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$.
4. Minimize $f(x) = x_1^2 - x_1x_2 + 3x_2^2$ starting at $x^0 = [1, 2]^T$
5. Minimize $f(x) = [1 + (x_1 + x_2 - 5)^2][1 + (3x_1 - 2x_2)^2]$ taking the initial points (10,10) and (2,2).

Newton Method in a Non-Quadratic Function

1. Minimize $f(x) = x_1^2 + 4x_2^2 - \cos(x_1 + x_2)$ with start point $\begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$.

$$\nabla f_1 = \begin{Bmatrix} f_{x_1} \\ f_{x_2} \end{Bmatrix} \text{ at } X_1 = \begin{Bmatrix} 2 + \sin(2) \\ 8 + \sin(2) \end{Bmatrix} \text{ at } (1,1) = \begin{Bmatrix} 2.91 \\ 8.91 \end{Bmatrix}$$

2. Gradient descent/ascent method

Gradient Descent is an iterative optimization technique used to find the minimum value of a function. It is widely used in Machine Learning, Neural Networks, Optimization problems, Curve fitting, etc.

The basic idea is: Move step-by-step in the direction of the negative gradient (steepest descent) to reach the minimum point.

Suppose we want to minimize a function $f(X)$. The gradient descent iterative formula is

$$X_{i+1} = X_i - \lambda_i \nabla f(X_i)$$

where

$$\lambda_i = \frac{[\nabla f(X_i)]^T \nabla f(X_i)}{[\nabla f(X_i)]^T [H_i] \nabla f(X_i)}, \quad [H_i] \text{ is the Hessian matrix.}$$

Problems:

Approximate the minimum point of the following functions near the given point using Gradient descent/ascent method (Perform three iterations).

1. $f(x, y) = 3x^2 + y^2$ near (1,3)



2. $f(x, y) = x^2 - xy + y^2$ near $(1, \frac{1}{2})$
3. $f(x, y) = 4x^2 - 4xy + 2y^2$ near $(0, 0)$
4. $f(x, y) = (\frac{3}{4}x - \frac{3}{2})^2 + (y - 2)^2 + \frac{1}{4}xy$ near $(5, 4)$
5. $f(x, y) = 4x^2 - 8xy + 6y^2$ near $(1, 1)$
6. $f(x, y) = x - y + 2x^2 + 2xy + y^2$ near $(0, 0)$

3. Fibonacci Method

The Fibonacci search is based on the sequence of Fibonacci numbers which are defined by the equations

$$F_0 = 1, F_1 = 1$$

$$F_{N+1} = F_N + F_{N-1}, \text{ for } N = 1, 2, \dots$$

Thus, the Fibonacci numbers are 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

Fibonacci algorithm

1. Determine the number of function evaluations, $N - 1$.
2. Divide the initial interval $[a_1, b_1]$ evenly into F_N subintervals and hence the length of each subinterval is $\frac{1}{F_N}(b_1 - a_1)$. Grid point P_i , $i = 1, \dots, F_{N-1}$, in the interval can be expressed as $P_i = a_1 + \frac{i}{F_N}(b_1 - a_1)$.
3. In the first iteration, take the F_{N-2}^{th} grid point from left as x_1^1 and the F_{N-1}^{th} grid point from left as x_2^1 , that is, $x_1^1 = \frac{F_{N-2}}{F_N}(b_1 - a_1) + a_1$ and $x_2^1 = \frac{F_{N-1}}{F_N}(b_1 - a_1) + a_1$

i.e.

$$x_1^1 = a_1 + \frac{F_{N-2}}{F_N}(b_1 - a_1) \quad \text{and} \quad x_2^1 = a_1 + \frac{F_{N-1}}{F_N}(b_1 - a_1), \quad k = 0, 1, 2, 3, \dots$$

Compare the function values at these two points $(f(x_1^1), f(x_2^1))$ and decide the new interval. If $f(x_1^1) > f(x_2^1)$ then the new interval is $[x_1^1, b_1]$ else the new interval is $[a_1, x_2^1]$.

4. Repeat the step 3 for other k values to obtain $x_1^2, x_2^2, \dots$. You will find that $N - 1$ function evaluations are made at $N - 1$ of the F_{N-1} grid points.
5. After completing all the prescribed iterations, determine the approximate value of x as the midpoint of the final interval obtained. The corresponding minimum value of the function is $f(x)$.

Note: If a minimizer has to be found with tolerance (level of uncertainty) ϵ , then the length of the last interval $\frac{b-a}{F_N} < \epsilon$.

Examples:

1. Apply Fibonacci search method to find the value of x that minimizes $f(x) = x^4 - 14x^3 + 60x^2 - 70x$ over the range $[0, 2]$ by performing 7 iterations.
2. Apply Fibonacci method to minimize $f(x) = x(x - 1.5)$ in $[0, 1]$ within the interval of uncertainty 0.25 of the initial interval.



3. Apply Fibonacci search method to find the value of x that minimizes $f(x) = (x - 3)^2 - 6x$ in the interval $[-5, 5]$ by taking 8 Fibonacci terms.
4. Use Fibonacci algorithm to minimize $f(x) = \frac{-1}{(x-1)^2} \left(\log x - \frac{2(x-1)}{x+1} \right)$, $\forall x \in [1.5, 4.5]$ by performing 7 iterations.

4. Sequential search

The sequential search (or linear search) algorithm is a simple and straightforward method used to find a specific element in a list. It works by checking each element in the list one by one until the desired element is found or the list ends.

Here's a mathematical formulation and a step-by-step description of the sequential search algorithm:

1. Input

- A list L of n elements: $L = [L_1, L_2, \dots, L_n]$
- A target value T

2. Algorithm

- Start with the first element of the list.
- Compare the current element L_i with the target value T .
- If $L_i = T$, the target value is found, and return the index i .
- If $L_i \neq T$, move to the next element and repeat the comparison.
- If the end of the list is reached and the target value is not found, return -1 or another indicator that the target is not in the list.

The 3-point search algorithm is a technique used for finding the maximum or minimum value of a unimodal function. It involves evaluating the function at three points and iteratively narrowing the search interval based on these evaluations. The algorithm works as follows:

1. Choose three points a , b , and c such that $a < b < c$.
2. Evaluate the function f at these points: $f(a)$, $f(b)$, and $f(c)$.
3. Determine the new search interval based on the following cases:
 - If $f(b)$ is less than both $f(a)$ and $f(c)$, then the new interval is (a, c) .
 - If $f(a)$ is less than $f(b)$, then the new interval is (a, b) .
 - If $f(c)$ is less than $f(b)$, then the new interval is (b, c) .
4. Repeat steps 1-3 with the new interval until the interval is sufficiently small.